Arista Access Node HSR-ACC-27 Hardware/Environmental Fault Root Cause Analysis and Resolution

Node: HSR-ACC-27 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_9931

Description

This Standard Operating Procedure (SOP) outlines the steps to be followed when a hardware/environmental fault occurs on the Arista Access Node HSR-ACC-27, located at HSR Layout Access. The failure domain is characterized by ASIC memory parity errors, line card crashes, and power supply redundancy loss, resulting in the ALARM_9931 code being raised. This scenario is specific to the Arista Access role, where the device serves as a gateway for customer traffic, providing Layer 2 and Layer 3 services. The underlying protocol and hardware mechanics responsible for this failure domain include the Arista Extensible Operating System (EOS), which runs on the device's ASIC-based hardware. The EOS is responsible for managing the device's hardware resources, including the line cards and power supplies. The ASIC memory parity errors occur due to hardware malfunctions, while line card crashes and power supply redundancy loss can be caused by environmental factors such as overheating, power surges, or equipment failure. It is essential to address these issues promptly to prevent network downtime and ensure high availability.

Pre-requisite Checks

- Verify the site location and ensure access to the HSR Layout Access site.
- Check the redundant link status and ensure it is functioning correctly.
- Notify the NOC and obtain permission to proceed with the troubleshooting and resolution process.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware, Expected Output: Hardware information, including ASIC and line card details., Description: This command will provide an overview of the device's hardware configuration, including ASIC and line card details. It will aid in identifying potential issues related to the hardware.

Command: show environment, Expected Output: Environmental information, including temperature and power consumption details., Description:
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```
This command will provide information on the device's environmental settings, including temperature and power consumption details. It will help identify potential environmental factors that may be contributing to the hardware fault.
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Resolution Steps

Step 1: Step: Power Cycle the Device, Command: reboot, Description: Power cycling the device may resolve the issue if it is related to a software or hardware glitch.

Step 2: Step: Replace the Faulty Line Card, Command: no module 1/1, Description: Replacing the faulty line card will resolve the issue if it is related to hardware failure.

Step 3: Step: Replace the Power Supply, Command: no power-supply 1, Description: Replacing the power supply will resolve the issue if it is related to power supply redundancy loss.

Post-Check Verification

```
Command: show hardware, Expected Output: Hardware information, including ASIC and line card details., Description: Verify that the hardware fault has been resolved by checking the hardware information.

Command: show environment, Expected Output: Environmental information, including temperature and power consumption details., Description: Verify that the environmental settings are within normal ranges.
```

Roll-back Procedures

- Step: Restore the Original Configuration, Command: copy startup-config running-config, Description: Restore the original configuration to ensure that any changes made during the troubleshooting process do not persist.

- Step: Reboot the Device, Command: reboot, Description: Reboot the device to ensure that any changes made during the troubleshooting process do not persist.